Knowledge Cutoff Effects on LLM-Generated Literature Reviews in Rapidly Evolving Fields

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Abstract

Large language models (LLMs) are increasingly used to support literature discovery, synthesis, summarization, and the preparation of research reviews. Their usefulness, however , is constrained by a fundamental temporal asymmetry: scientific knowledge evolves continuously, whereas a conventional pretrained model acquires most of its parametric knowledge from a finite historical training corpus. This temporal mismatch is particularly consequential in rapidly evolving fields such as artificial intelligence, biomedicine, climate science, cybersecurity, and emerging technologies, where important findings, terminology, datasets, methods, and consensus positions may change substantially within months. This paper examines how knowledge cutoffs affect LLM-generated literature reviews and argues that the problem extends beyond simple omission of recent papers. Cutoffs can produce systematic distortions in literature coverage, citation selection, interpretation of research trends, assessment of methodological consensus, and identification of research gaps. The paper synthesizes research on temporal knowledge in LLMs, retrieval-augmented generation (RAG), continual learning, knowledge editing, hallucination, citation reliability, and automated scientific literature synthesis. Current approaches—including web retrieval, scholarly RAG, temporal filtering, continual updating, model editing, agentic research systems, and citation verification—substantially reduce some cutoff effects but do not eliminate temporal misalignment. In particular , retrieval systems can themselves contain stale or conflicting information, while LLMs may fail to distinguish historical from current evidence. Recent benchmarks demonstrate persistent weaknesses in temporal awareness and outdated-information handling, while emerging systems such as OpenScholar demonstrate the potential of specialized retrieval and iterative verification. The paper identifies a central research gap: literature-review systems require evaluation not merely for factuality and citation correctness, but for temporal completeness, temporal attribution, trend fidelity, and reproducibility. Future systems should therefore treat time as a first-class dimension of scholarly evidence, combining timestamp-aware retrieval, source provenance, temporal reasoning, claim-level citation verification, uncertainty estimation, and human oversight.

Keywords: large language models; knowledge cutoff; literature review; scientific literature synthesis; temporal knowledge; retrieval-augmented generation; citation hallucination; continual learning; research agents; temporal alignment

1 Introduction

Literature reviews are fundamental to scientific research because they organize existing evidence, identify methodological developments, expose disagreements, and establish the rationale for new investigations. Traditionally, preparing a high-quality review requires iterative database searching, screening, reading, comparison, citation management, and expert synthesis. The growth of scientific publication has made this process increasingly difficult for individual researchers. Recent work on

automated literature research explicitly identifies the volume of scientific publications as a major bottleneck for human researchers and 1 motivates LLM-based review-generation systems as a means of reducing this cognitive burden (Wu et al., 2025). LLMs have changed this workflow by enabling researchers to ask questions in natural language and receive coherent summaries, comparisons, explanations, and draft reviews. However , conventional LLMs are not continuously synchronized with the scholarly record. Their parametric knowledge is primarily acquired during pretraining, and therefore a model with a training cutoff cannot inherently know papers, discoveries, terminology, or empirical results that emerged afterward. Retrieval-augmented generation (RAG) was introduced partly to address this limitation by combining parametric model knowledge with an external non-parametric memory that can be updated independently of model weights ([9]). The knowledge-cutoff problem is more complicated than simply asking whether an LLM "knows" a recent paper . Cheng et al. (2024) demonstrate that the effective cutoff of an LLM can differ from its officially reported cutoff because training corpora contain documents with different temporal properties. Older material can remain present in newer web crawls, while reduplication procedures can create additional temporal inconsistencies. Thus, a model's temporal knowledge should be understood as a distribution rather than as a single date. For literature reviews, this distinction is particularly important. A review is not merely a collection of facts. It is a structured representation of how a field has developed. If an LLM misses recent studies, it can misrepresent the current state of evidence, incorrectly identify an old problem as unsolved, fail to recognize a newly established method, overstate the importance of obsolete techniques, or describe a historical research debate as if it were current. Consequently, knowledge cutoff effects can become epistemic distortions rather than ordinary factual errors. This paper examines these effects systematically. It focuses on five questions: How do knowledge cutoffs influence LLM-generated literature reviews? Why are rapidly evolving fields especially vulnerable? Which technical approaches currently mitigate temporal limitations? What new failure modes arise when retrieval and agentic systems are introduced? What evaluation and architectural principles are required for temporally reliable literature-review systems? The paper is a conceptual and narrative synthesis of scholarly research available through August 2026. It does not claim to constitute a systematic review of every publication on LLM-assisted literature review. Instead, it integrates research from LLM temporal reasoning, RAG, continual learning, scientific literature synthesis, hallucination, citation verification, and research-agent evaluation.

2 Current Approaches and Developments

2.1 Retrieval-augmented generation

RAG is the most established solution to knowledge staleness. Instead of relying entirely on parametric memory, the system retrieves external documents and conditions generation on them ([9]). For literature reviews, a basic RAG pipeline can be represented as: where is the research question, is the external corpus, is retrieval, is retrieved context, and is generation. RAG allows the underlying LLM to remain unchanged while the evidence corpus is updated. However , retrieval quality becomes a critical dependency.

2.2 Scholarly RAG

General web search is not necessarily sufficient for scholarly review generation. Scientific systems need metadata, abstracts, full text, publication dates, versions, citations, author information, and preferably persistent identifiers. Q→R(Q,D)→C→G(C,Q) Q D R C G 7 PaperQA represents

an early research-oriented RAG approach, retrieving full-text scientific literature and synthesizing evidence with citations ([10]). OpenScholar represents a more recent development. [2] introduced a specialized retrieval- augmented language model backed by a datastore containing approximately 45 million open-access papers. The system combines specialized retrieval, reranking, and iterative self-feedback. Its evaluations found substantial improvements over conventional LLM-only approaches and strong performance in citation accuracy and scientific-answer correctness. The significance of OpenScholar is not simply higher retrieval accuracy. It demonstrates that architecture and evidence infrastructure can be as important as model scale for scientific synthesis.

2.3 Temporal filtering

A simple but powerful strategy is to make time explicit during retrieval. A query can specify: For a "state of the field as of 2023" review, documents published after 2023 should not be retrieved. Conversely, a current review should prioritize documents close to the present. Temporal filtering can be combined with relevance: where: = semantic relevance; = temporal relevance; = source authority; = citation or scholarly significance. Such scoring prevents a highly cited but obsolete paper from automatically outranking a recent and directly relevant study.

2.4 Continual learning

Another approach is to update model parameters periodically. Continual-learning research distinguishes continual pretraining, domain-adaptive pretraining, and continual fine-tuning ([15]; Shi et al., 2024). The advantage is that frequently needed new knowledge becomes part of the model's internal representation. The disadvantage is cost and the risk of catastrophic forgetting. For literature-review systems, continual training also introduces a reproducibility problem: if a model changes continuously, two researchers querying the "same" system at different times may obtain different reviews. D =t {d D publication_date(d) t} Score(d)= R(d,q)+ T(d,t)+ A(d)+ C(d) • R • T • A • C 8 Thus, continuously updated models may improve freshness while reducing experimental reproducibility.

2.5 Knowledge editing

Knowledge editing modifies specific model representations rather than retraining the entire system. Temporal knowledge editing is especially relevant when facts become outdated. [16] argue that outdated knowledge should not necessarily be erased. Historical information may remain valuable when a user asks a historical question. Their work distinguishes correcting intrinsic model errors from integrating new knowledge that supersedes previous knowledge. This distinction is essential for literature reviews. A good review must preserve the history of a field rather than simply overwrite old knowledge.

2.6 Fact-duration modeling

[17] propose predicting how long facts remain valid. Their results indicate that volatile facts should trigger stronger reliance on external sources. A literature-review system could extend this principle from facts to claims. For example: established mathematical theorem → low temporal volatility; drug efficacy estimate → medium/high volatility; state-of-the-art benchmark result → high volatility; current regulatory guidance → very high volatility. This suggests a promising design principle: retrieval effort should be proportional to expected temporal volatility.

2.7 Agentic deep research

Recent systems increasingly use agents that plan searches, execute multiple retrieval steps, compare sources, and generate structured reports. DeepResearch Bench evaluates such systems using 100 PhD-level tasks spanning 22 fields and emphasizes both research quality and citation accuracy. Agentic architectures can reduce cutoff effects because they can access information published after the model's training cutoff. However , they also introduce additional failure modes: query drift; source-selection bias; retrieval loops; incomplete search coverage; premature stopping; inconsistent source ranking; opaque decision processes; and temporal contamination. • • • • • • • • • • • • 9 Consequently, "agentic" does not automatically mean "systematic."

2.8 Citation verification

Citation verification is an increasingly important safeguard. [13] demonstrate that LLM-generated literature reviews can contain fabricated references, while recent research continues to document reference hallucination in academic-style outputs. Verification should occur at multiple levels: Existence: Does the paper actually exist? Metadata: Are authors, title, year , venue, and identifier correct? Temporal validity: Was the paper available by the relevant cutoff? Entailment: Does the cited paper support the claim? Scope: Does the cited evidence support the strength of the wording? Completeness: Are major competing or contradictory studies missing? OpenScholar's evaluation illustrates the value of treating citation accuracy as a first-class metric rather than an afterthought.

3 Research Challenges and Limitations

3.1 No universal definition of "current"

A literature review can be current relative to: the publication date of the review; the last database search; a specified historical cutoff; the latest peer-reviewed literature; the latest preprints; a field-specific guideline; or a dynamically changing evidence base. Therefore, temporal correctness requires an explicit reference time.

3.2 Preprints and publication versions

Rapidly evolving fields often rely heavily on preprints. A paper may exist as a preprint in January, a conference paper in June, and a journal version the following year . A temporally accurate system must represent these as related versions rather than independent evidence. 1. 2. 3. 4. 5. 6. • • • • • • • 10

3.3 Retractions and corrections

Scientific knowledge is not monotonically additive. New papers can invalidate earlier results through corrections, retractions, failed replications, or improved methodologies. A simple "latest paper wins" strategy is therefore inadequate. Temporal reasoning must be combined with evidence quality and scientific authority.

3.4 Retrieval bias

RAG systems inherit biases from their retrieval corpus. If the corpus underrepresents certain journals, languages, geographic regions, disciplines, or publication types, the generated review will

inherit those gaps. This means that retrieval can improve freshness without improving representativeness.

3.5 Conflicting evidence

Scientific fields rarely progress through simple replacement. Competing studies may coexist for years. An LLM-generated review may collapse disagreement into a single narrative because language generation rewards coherence. This creates a tension between readability and epistemic fidelity.

3.6 Citation correctness versus citation completeness

A review can have 100% valid citations and still be poor if it omits major recent studies. This suggests that evaluation should distinguish: from Both are necessary for scholarly reliability.

3.7 Evaluation benchmarks remain incomplete

Recent benchmarks such as EvolveBench and HoH are significant advances, but most evaluations still focus on question answering rather than full literature-review generation. A literature review requires broader evaluation dimensions: search completeness; chronological coverage; citation accuracy; CitationAccuracy EvidenceCoverage. • • • 11 claim-evidence alignment; synthesis quality; disagreement representation; methodological comparison; gap identification; and reproducibility.

3.8 Reproducibility

Dynamic retrieval creates a paradox. A system can be more current but less reproducible. If the same query is executed six months apart, the retrieved literature will differ . Therefore, scholarly use requires preservation of: query strings; search date; database version; retrieved document IDs; document versions; ranking results; model version; prompts; and generation settings. Without this information, an AI-generated review cannot easily be audited.

4 Research Gaps and Future Directions

4.1 Temporal completeness as a formal evaluation metric

Current evaluation commonly emphasizes factual correctness and citation accuracy. Future literature-review benchmarks should measure temporal completeness. One possible metric is: where if a substantively important publication or finding is represented and reflects its relevance or importance. This would distinguish a review that is factually correct but incomplete from one that provides broad current coverage. • • • • • • • • • • • • • • • TC= w i=1 n i wI i=1 n i i I =i 1 w i 12

4.2 Claim-level temporal provenance

Every substantive claim should ideally have a provenance tuple: This would enable a system to state not simply "Method X is effective," but: "Studies published between 2024 and 2026 generally report improved performance under conditions A and B, although evidence remains mixed under condition C." Such statements better represent scientific uncertainty and temporal evolution.

4.3 Temporal knowledge graphs

Future systems could represent literature as a temporal graph: where nodes represent papers, methods, datasets, claims, and findings; edges represent citations, comparisons, replications, contradictions, and extensions; and records temporal validity. This would allow the LLM to reason over changes in relationships rather than merely retrieve documents.

4.4 Temporal conflict resolution

Systems need explicit mechanisms for deciding what to do when sources disagree. A useful hierarchy might consider: publication date; peer-review status; methodological quality; replication evidence; retraction/correction status; sample size; domain authority; and directness of evidence. The latest source should not automatically dominate.

4.5 Time-aware research agents

Future deep-research agents should maintain a temporal state during search. Rather than simply accumulating sources, the agent should ask: When was this claim established?

P=(source,date,version,evidence,confidence) G=(V,E,T) T 1. 2. 3. 4. 5. 6. 7. 8. • 13 Has it subsequently changed? Is there newer contradictory evidence? Was this paper available by the requested cutoff? Is this a preprint or final publication? Has this result been replicated? This would transform research agents from document collectors into temporal evidence synthesizers.

4.6 Benchmarking literature reviews rather than individual answers

The field needs benchmark suites specifically designed for AI-generated literature reviews. Such benchmarks should contain rapidly changing topics with known publication timelines and expert-created reference reviews. Evaluation should measure: Dimension Core question Temporal coverage Did the system find important recent work? Citation validity Do cited papers exist and have correct metadata? Citation entailment Does each citation support the claim? Temporal attributionIs evidence placed in the correct historical context? Trend fidelity Does the review accurately describe field evolution? Conflict representationAre competing findings represented? Gap validity Are claimed gaps actually unresolved? Reproducibility Can another researcher reconstruct the review?

4.7 Human-AI collaboration

The evidence increasingly supports treating LLMs as research assistants rather than autonomous replacements for scholarly judgment. A 2025 scoping review concluded that LLM applications in systematic reviews are promising but not yet sufficiently validated for unrestricted use. Human researchers remain particularly important for: assessing methodological quality; interpreting contradictory findings; judging novelty; identifying disciplinary context; evaluating source credibility; and determining whether a research gap is genuinely meaningful. The optimal workflow is therefore likely to be human-directed, machine-accelerated synthesis. • • • • • • • • • • • • 14

4.8 Explicit uncertainty and abstention

An LLM should be able to say: "The available evidence after 2025 is insufficient to establish a consensus." This is preferable to generating a confident synthesis from incomplete evidence. Recent research on prompting earlier knowledge cutoffs shows that models can be instructed to simulate temporal restrictions but struggle to prevent leakage from causally related post-cutoff information

(Gao et al., 2025). This demonstrates why temporal awareness cannot depend exclusively on prompting. The system must have mechanisms for verifying temporal admissibility.

5 Conclusion

Knowledge cutoffs create a fundamental reliability problem for LLM-generated literature reviews in rapidly evolving scientific fields. The problem is not limited to missing recent papers. A static knowledge boundary can influence the perceived state of the field, distort citation patterns, preserve obsolete consensus, generate false research gaps, and misrepresent emerging methodological trends. The central insight is that scholarly synthesis is inherently temporal. A literature review describes not only what is known, but when it became known, how evidence changed, which findings were superseded, and where uncertainty remains. Conventional LLMs are optimized primarily for linguistic prediction and general reasoning, not for maintaining an auditable temporal model of scientific knowledge. RAG provides an important architectural response by separating language-generation capabilities from external knowledge storage. Scholarly systems such as PaperQA, LitLLM, automated review-generation pipelines, and OpenScholar demonstrate increasingly sophisticated ways of combining retrieval with LLM- based synthesis. Nevertheless, retrieval introduces its own temporal problems: outdated documents can coexist with current ones, citation networks can privilege older literature, and retrieval systems can fail to distinguish different versions or conflicting findings. The emerging research on temporal benchmarks reinforces this conclusion. EvolvingQA, EvolveBench, and HoH demonstrate that temporal awareness and outdated-information handling remain unresolved even for highly capable models. Future literature-review systems should therefore treat time as a first-class property of evidence. Reliable systems should combine timestamp-aware scholarly retrieval, temporal provenance, claim-level citation verification, source-version tracking, conflict resolution, uncertainty estimation, and reproducible search logs. Evaluation should expand beyond factuality toward temporal completeness, trend fidelity, gap validity, and historical attribution. 15 The ultimate goal should not be an LLM that merely produces a review that sounds current. It should be a research system that can demonstrate why its review is current, which evidence supports each claim, when that evidence was available, what evidence contradicts it, and what remains unknown. Such capabilities would transform LLM-generated literature reviews from fluent summaries into auditable, temporally grounded instruments for scientific knowledge synthesis.

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